C. Chessboard

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: bitmasks, brute force, implementation | https://codeforces.com/problemset/problem/961/C

Magnus decided to play a classic chess game. Though what he saw in his locker shocked him! His favourite chessboard got broken into 4 pieces, each of size n by n, n is **always odd**. And what's even worse, some squares were of wrong color. j-th square of the i-th row of k-th piece of the board has color $a_{k, i, j}$; 1 being black and 0 being white.

Now Magnus wants to change color of some squares in such a way that he recolors minimum number of squares and obtained pieces form a valid chessboard. Every square has its color different to each of the neightbouring by side squares in a valid board. Its size should be $2n$ by $2n$. You are allowed to move pieces but **not allowed to rotate or flip them**.

Input

The first line contains **odd** integer n ($1 \le n \le 100$) — the size of all pieces of the board.

Then 4 segments follow, each describes one piece of the board. Each consists of n lines of n characters; j-th one of i-th line is equal to `1` if the square is black initially and `0` otherwise. Segments are separated by an empty line.

Output

Print one number — minimum number of squares Magnus should recolor to be able to obtain a valid chessboard.

Examples

Input
1 0 0 1 0

Output
1

Input

```
3
101
010
101

101
000
101

010
101
011

010
101
010
```

Output

```
2
```